

The Viable System Model

An Introduction to Theory and Practice

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Historical development

Firmly grounded in the theories of cyberneticians McCulloch and Ashby on neural networks, self-organization and the Law of Requisite Variety, Beer pioneered Organizational Cybernetics '*the science of effective organization*'. Centrally relevant is the Viable System Model (VSM) - a systemic tool to support organizational (re) design. Beer [1–3] provided the background theory and guidance on its application, which he originally tested in the steel industry in Sheffield (UK), and further developed during the Cybersyn project, redesigning the Chilean industrial economy in the 1970s [2].

Beer defined a *viable system* as a system 'capable of independent existence'. Focusing on identifying patterns of communication between organized teams in a social organization, he identified the **invariances** which apply universally, regardless of the size or nature of the organization. Such 'invariances', are explained in Axioms, Laws and Principles of Organisation, which can be applied recursively to model organizations of any scale and complexity. The VSM defines the necessary and sufficient conditions for organizational viability [4]. It offers criteria for designing organizations capable of co-evolving with continuously changing environments, by developing fractal structures focused on effective self-organized operational teams, serviced by meta-systemic management. The model enables the practitioner to map the 'warts-and-all' reality of the way an organization works, and then assess it using VSM criteria of viability.

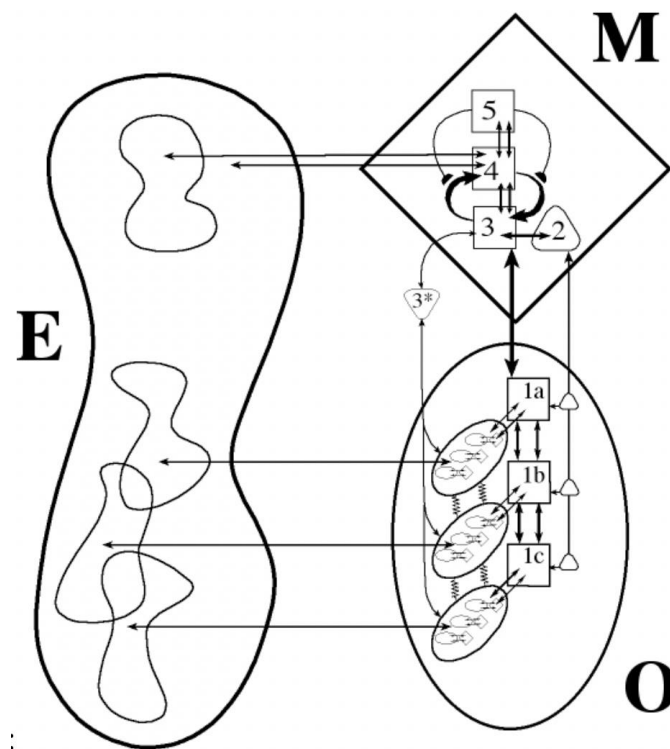
Although the theory was developed long ago, it is only now in the 21st century, when organizations are globally interconnected, and operating in more fluid and complex environments, that the need for operating in flatter and more networked structures has been recognised. It is not surprising that the VSM has been recently re-discovered and increasingly used both in academia and in practice. Well known approaches to use the VSM include Espejo & Harnden [5]; Espejo *et al.* [6]; Yolles [7]; Walker [8]; Espinosa [9]; Schwaninger [10]; Christopher [11], Hoverstadt [12]; Leonard [13]; Espejo & Reyes [14]; Perez-Rios [15], Espinosa & Walker [16] and Espinosa [27]. It has been applied at all scales from small work-groups to subject matter experts (SMEs), communities and nation states, and experienced practitioners often tell of the power and speed of the model to identify and solve issues of organizational structure.

The Viable System Model

Beer modeled an organization as a neural network, where each node is itself an autonomous, viable system, and all nodes together constitute a purposeful and cohesive organization. He used a meta-language which describes the basic components and interactions of such a network: it includes three types of components:

1. the Operational units (O) of the organization - those directly responsible for implementing the organizational purpose;
2. the environment (E) – the niche to which the organization is structurally coupled and with which it co-evolves; and
3. the Meta-system (M) – the managerial and technical support required to coordinate the operational nodes and provide them with the resources, technology and knowledge to effectively implement their tasks.

A simple VSM is depicted in Figure 1:



The Operational ellipse is designated **System 1** and it is itself a cluster of smaller interacting operational units: in the above example it is composed of three smaller operational units (labeled 1a, 1b, and 1c) – each of which has its own operational management – represented by a small diamond connected to the operation. Each System 1 has its own operational niche, represented by a small cloud shape on the left-hand side, which includes all stakeholders they interact with while implementing their tasks, such as

customers, suppliers, competitors and interveners. The big cloud shape integrates all the local operating environments.

The meta-systemic management is represented by a diamond: it is composed of four subsystems (designated systems 2, 3/3*, 4 and 5), which provide the services to the operational elements so that they cohere into a Viable System. The various arrows represent a myriad of loops of often highly complex interactions between the five systems and the environment in which the viable system operates. In contrast to traditional linear thinking and hierarchical relationships, in the VSM, everything works in loops: discussions take place continuously within the various operational teams, supported when required by managers, to improve the teams' learning on the best ways to jointly implement specific tasks. Decisions are made which result in actions, and teams continuously monitor and assess processes and results, and critically review their learning. Deviations from plans are noted and considered, plans are adjusted and new actions undertaken.

Each loop represents a relationship, and management is seen as a continuous 'coaching' role which ensures on-going self-assessment and adjustment. Ideally, all information should be in real time to enable instantaneous response, in the same way we react when our hand touches a hot object. Rather than allocating blame, the VSM illustrates the way the operational parts of a business interact with each other and with their trading environment, and how management coordinates to cohere, adapt and to ensure effective organization.

The Five Systems.

System 1 is the Operational team producing core products or services which bring to reality the identity of the viable system to which they belong. It is rightly and properly focussed on managing the complexity of its interactions with its local environment. It has to continuously learn how to manage the variety of its niche, in order to be able to adapt to changes and emerging challenges.¹

System 2 deals with the inevitable conflicts of interests which emerge as the operational units interact – e.g. competition for resources. Beer describes its role as “damping oscillations”: without a System 2, the organizational system would shake itself to pieces. System 2 is there to harmonize Systems 1's interactions, to keep the peace, to prevent and deal with their conflicts of interest. Examples of Systems 2 are operational rules, organizational values and codes of conduct, process standards, schedules, and information systems.

System 3 is primarily concerned with facilitating synergy and the self-regulation of the operational units. It looks at the entire interacting cluster of operational units from its meta-systemic perspective and considers ways to maximize its effectiveness through fostering operational self-management and promoting collaboration. Beer talks about an “*explosion of potential*” which emerges from collaboration

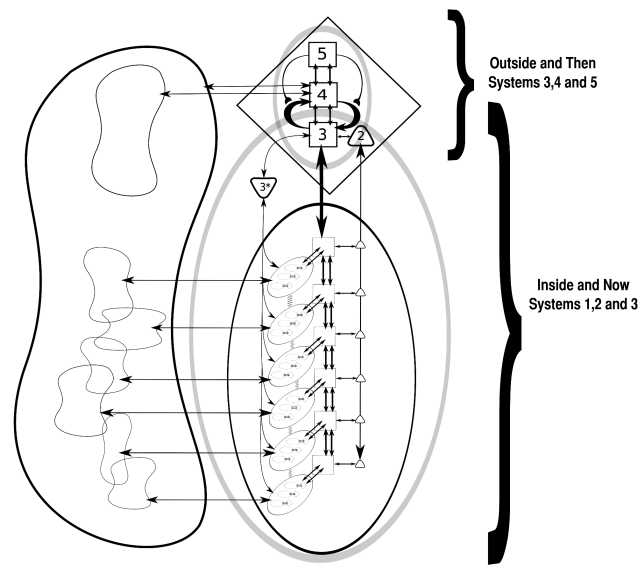
¹ We follow Maturana and Varela's understanding of the 'niche,' as the one which emerges from recurrent interactions with the agents, institutions or organizations to which the organization is structurally coupled [17].

and symbiotic relationships between the Systems 1. System 3 functions include recurrent conversations between general and operational managers on resource negotiation (i.e. financial, human, technological and infrastructure), accountability, and corporate norms. *System 3** is a particular case of System 3 responsible for providing alternative, sporadic and complementary information to meta-systemic managers about the happenings at the operational level (e.g. sporadic auditing).

While Systems 1 to 3 jointly provide the organizational structures to deal with the '*inside*' and '*now*' of the organization, it is only by being attentive to the challenges in the external environment that the organization would be able to continuously adapt. In order to do so, it needs to explore its '*outside*' and '*then*' on a continuous basis. This is precisely the role of *System 4*: to ensure the whole system can adapt to a rapidly changing and potentially hostile environment. It scans the outside world, looks for threats and opportunities, undertakes research, and proposes innovations. It develops and proposes the best opportunities and scenarios for organizational, people, market and product development and clarifies the various possible pathways it could follow. Without a System 4, the system would be unable to adapt to the complexity of the external environment in which it operates.

System 5 defines and develops identity, policy, and provides closure for the organizational system. It develops the vision, and ground rules of the organization, and the policies that knit the organizational social capital. Beer [1] states that System 5 puts the rules into practice, by creating a corporate ethos - an atmosphere. System 5 is also responsible for creating a balanced decision-making context, informed by multiple perspectives, including a balanced representation from Systems 3 and 4, which counterbalance and complement each other. Without a System 5, there would be no shared organizational identity and development of social capital, and fragmentation would be inevitable.

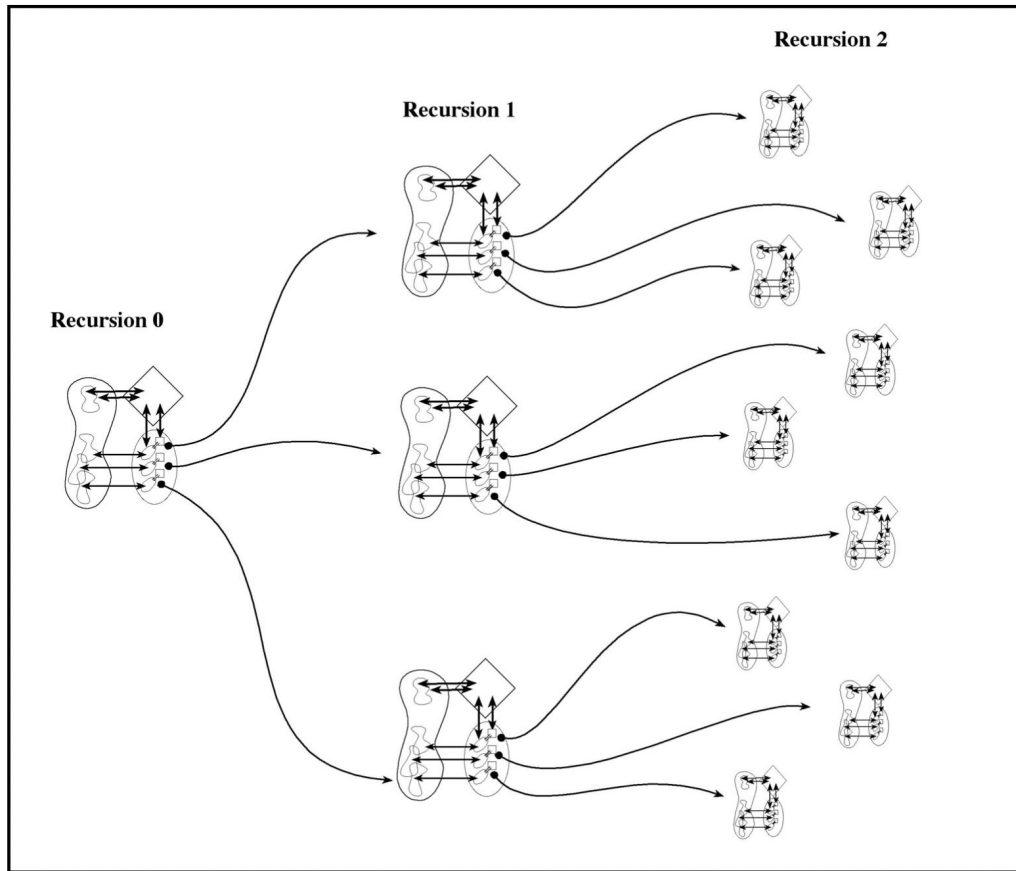
These five subsystems are the building blocks which are used to construct a viable system. The VSM also clarifies the pattern of relationships which specify the intricate interconnections between the five subsystems and their niches: the pattern of relationships which define the day-to-day, here-and-now workings of the enterprise (Inside and Now composed of Systems 1 2 and 3), and the way decisions are made to ensure the whole system adapts to a changing environment (Outside and Then, composed of Systems 3,4 and 5), as shown in Figure 2.



Managing organizational complexity

When modeling a system to study its behavior, we identify some key ‘variables’ and observe their behavior (numerical values or qualitative observations). [We talk about ‘the state’ of a system as the set of (numerical or qualitative) values which its (*observer chosen*) variables have at that instant. We can define ‘*Complexity*’ as ‘all the possible states that a system can exhibit at a particular moment’; and ‘*Variety*’ as the number of possible states *observed by an observer*.] The viewpoint of the observer determines the choice of variables, the way of measuring them, and therefore the observed complexity of a system. This was clearly explained by both Ashby [18] and Beer ([1], p.32-48) and clearly corresponds to a constructionist ontology.

Beer explained that the operational units need to be able to cope with their environmental variety (V_e in Figure 3), which is normally far higher than the organizational or task variety (V_o in Figure 3). The variety of the operational management is orders of magnitude smaller than the operational team’s variety. Ross Ashby’s ‘Law of Requisite Variety’ is illustrated in figure 3.



This Law explains why managers always operate facing uncertainty and dealing with limited amounts of information and knowledge; management decision making has to deal with navigating uncertainty and ambiguity. Beer explains that any manager, in principle, is unable to deal with the variety of one subsidiary System 1 (never mind all the subsidiary systems) ([3], p. 37).

Managers, operational teams and organizations, in general are continuous learners. Understanding it helps to design structures, processes, and technology prepared to cope with the natural imbalances in complexity and variety between them, their operational teams, and their environment. The VSM offers detailed criteria on how to generate a context enabling better individual and team capabilities to emerge, by managing complexity more effectively. Beer [3] explained that an organization maintains its internal homeostasis, if it maintains its identity and reputation through balanced patterns of interactions with its niche, (remains structurally coupled) despite the system having to cope with the enormous variety of an unpredictable niche.

Inside and Now: Systems 1, 2 and 3.

System 1, the Operation, is responsible for doing the work, providing the goods or services, which ensure the organizational system implements its identity. This is the horizontal axis of the VSM. In order for a

System 1 to have ‘requisite variety’ it needs as much autonomy as possible to make its own decisions ‘in real time’ to respond promptly to the high variety demands of its niche: any theory of autonomy, of decentralization, lies in this simple fact.

However, the meta-system is charged with ensuring all the Systems 1 cohere by sharing resources and knowledge to maximize effectiveness, and thus must have a measure of control which will constrain the autonomy of the Systems 1. Again, this is a pattern of relationships: every System 1 is interacting with suppliers and customers; System 2 interacts with the Systems 1 to avoid conflicts of interest: System 3 continuously offers support and looks for synergies.

There is, consequently, a conflict: operational autonomy must be maximized to cope with its niche but constrained to ensure system cohesion. In order to resolve this issue, Beer proposed that Systems 1 autonomy is constrained *only* when the Meta-system needs to intervene in order to ensure system cohesion. This creates a positive dynamic balance: the operational units are given as much autonomy as possible which is only limited by the needs of System 2 to deal with conflicts or of System 3 to generate synergies.

Outside and Then: System 3, 4 and 5.

System 4 is concerned with the Outside and Then. It formulates plans in the context of both the outside world and its intense interaction with System 3, which ensures all plans are grounded in the knowledge of the capabilities of the organization. System 5 monitors Systems 3 and 4 to ensure all plans are within policy guidelines. Systems 3, 4 and 5 are the adaptation homeostat for the organization – see Figure 2.

The large arrows connecting Systems 3 and 4 portray a regular, thorough, intense interchange of ideas between these two systems. System 4 – which has a creative open- minded view - has an idea, works it up into a proposal and checks with System 3 – which has a practical, down to earth perspective. They counterbalance each other while developing an action plan which is both creative and feasible. System 5 monitors the decision-making process at the S3-S4 homeostat and intervenes if ethos or policy issues are being ignored. Information is designed to flow throughout the structure in real-time, binding together the various roles and perspectives. Team Syntegrity [19], deals with the massive complexity of large group’s decisions -in the 3/4/5 homeostat.

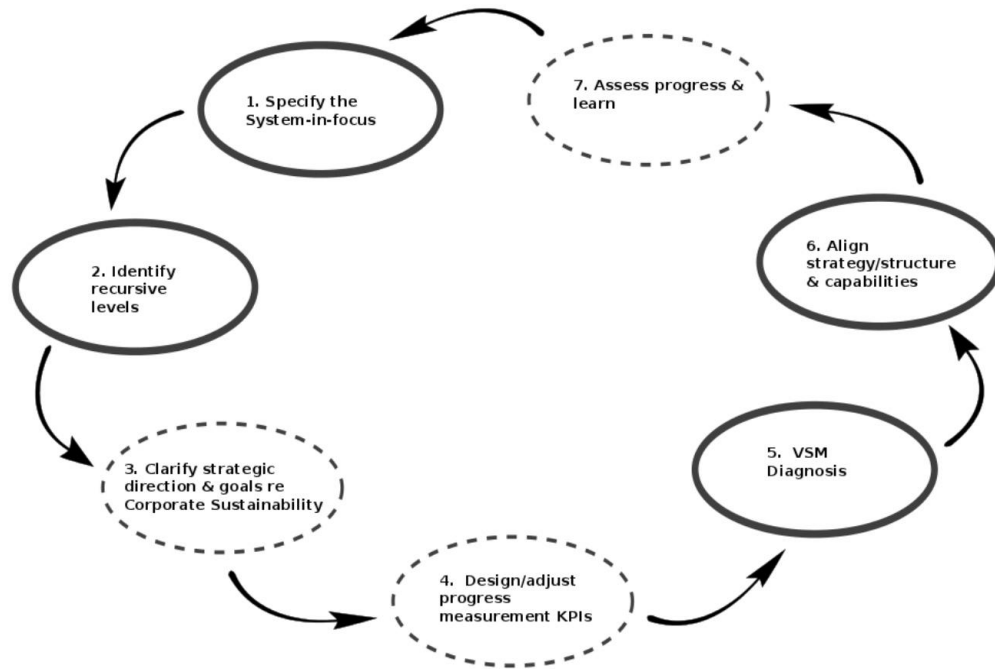
The Recursive Viable Systems Theorem

Beer explained that in a viable system each System 1 is in itself a viable system, which means that each operational unit must have in itself systems 1 to 5 - with its own operational management doing meta-systemic management. In his words:

‘In a recursive organizational structure, any viable system contains, and is contained in, a viable

system' ([1], p. 118, [3], p. xi).

In Figure 4, the Viable system in recursion 0 has an Operation which is composed of (e.g. three) viable systems; and each of these is also composed of (e.g. three) viable systems. (in real life the numbers will always vary). For effective functioning, each one of these 13 viable systems must be able to demonstrate that their five subsystems are properly designed, and that their “intricate interactions” are all functioning correctly, in as close to real time as possible. The elegance of this arrangement becomes apparent when looking at real life case studies, as the one presented in the next sections.



VSM Diagnosis

The VSM is a meta-language that can be used to facilitate collective learning among organizational teams or units about their effectiveness in implementing the organizational primary tasks (1st order learning); and also handling organizational complexity (2nd order learning). Figure 5 provides our ‘Self Transformation Methodology’ (STM) ([16], pp.95–99, 473–484; [26], pp.63-103) which suggests the use of the VSM in a participatory action research mode. This facilitates the development of shared models about the organization and its processes, and the identification of pathological or ineffective patterns of interactions among the different roles and units. The following case study summarizes a recent PhD action research project using the VSM in a Mexican SME ([20]).

Case study: Self transformation towards sustainability in Max Clean (2014-2015)

Max Clean is a 1,200 employee Mexican firm which offers industrial cleaning services in 6 cities. It has a turnover of US \$10 million a year having exhibited continuous growth for several years. Nevertheless, the challenges it was facing in 2014 were considerable including shortages of suitable labor, hazards in the surrounding communities, such as drug addiction, domestic violence, very low literacy levels, and uncontrolled levels of water contamination. At this point they requested advice from Martinez-Lozada about how to improve their structure and strategy to achieve more viability and sustainability. She led a participatory action research project for a year, adapting and enhancing the self transformation method (STM).

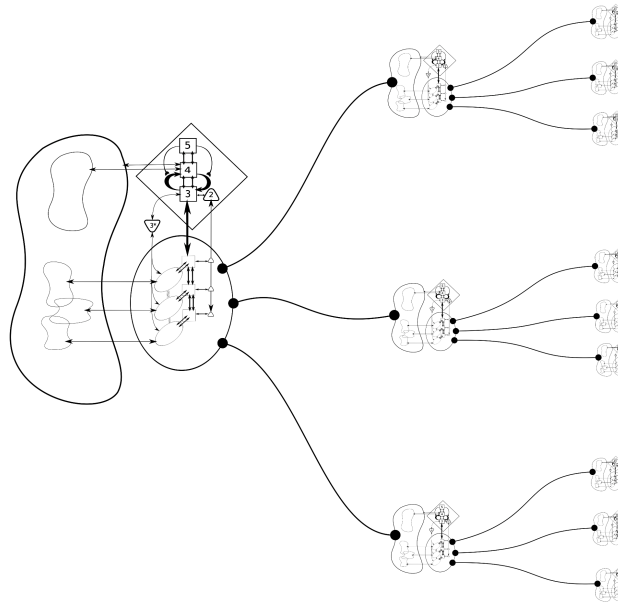


Table 1 below presents a summary of the stages and dates where the adjusted methodology was implemented.

<i>Stage</i>	<i>Activities</i>	<i>Date</i>
<i>Specify the system of interest</i>	Achieve agreement on the perceived organizational identity and purpose.	<i>Early</i>
<i>Determine the recursive levels</i>	Identify the nested viable systems existent in the firm and responsible for primary activities at different levels.	<i>September 2015</i>
<i>Clarify strategic direction and goals regarding corporate</i>	With the DC members, review the main strategic direction of the company, economic goals as well as the goals related to the social and environmental	September 2014

<i>sustainability</i>	dimensions. Study possible connections between goals.	
<i>Design / adjust progress measurement system</i>	Design the progress measurement system based on the essential variables of the system identified.	<i>September 2014</i>
<i>Structural and capabilities analysis and diagnosis</i>	At each recursion level, the VSM meta-questions are used to identify: • Structural problems within System 1 • Limitations: in the capacity to deal with core issues related to sustainability, • The existence of adapting and learning capabilities	<i>October –November 2014</i>
<i>Strategy, capabilities, structure alignment</i>	Analyze required organizational adjustments to achieve the implementation of organizational strategy	<i>November 2014</i>
<i>Achieve agreement regarding organizational changes</i>	Agree on organizational changes and suggest action plans to achieve the self-transformation process	<i>November 2014 – Early January 2015</i>
<i>Learning from progress measurement system</i>	Review advances in the implementation of action plans while at the same time check the indicators of the progress measurement system.	<i>January -March 2015</i>

Table 1:
Stages
of the

Methodology to support the implementation of corporate sustainability projects.

Martinez-Lozada began the VSM intervention in August 2014. During the first month she studied the company, its structure, strategy, vision and processes, through several meetings held with top management and HR personnel. A project team and a Diagnostic Committee (DC) were set up, who were responsible for guiding the company through the learning stages. It was clear that some of the stages required more than one iteration to deepen the learning. During the intervention, data was collected through semi-structured interviews with employees from different organizational levels and roles. Tools like agreeing on identity, recursive mapping, semi-structured interviews, workshops, flowcharts and mind maps were used to support the VSM diagnosis. There follows a summary of the stages of the methodology.

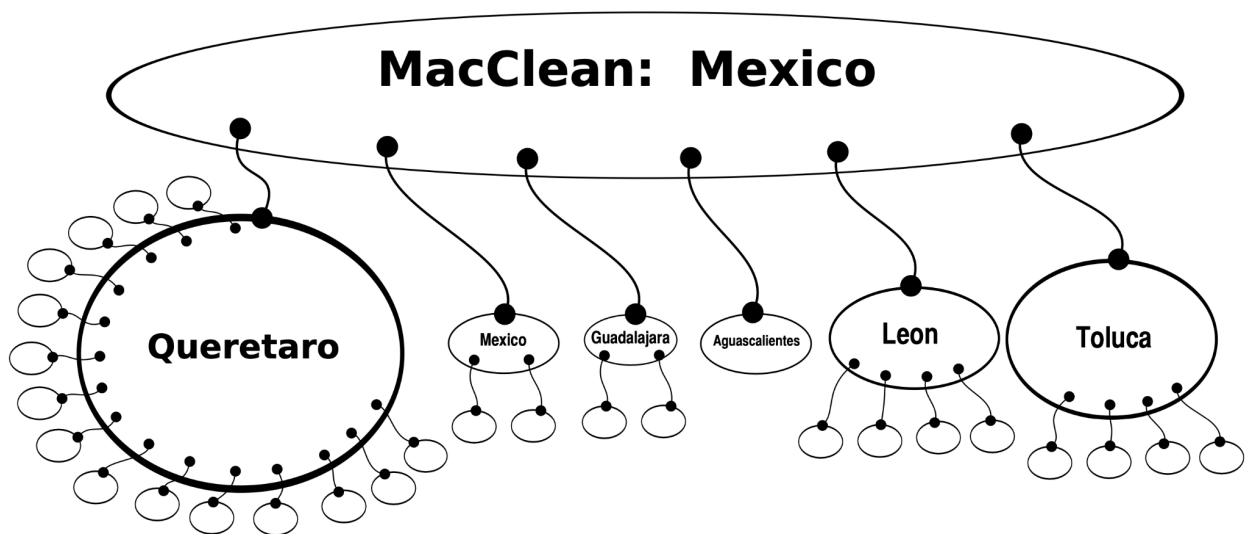
Agreeing on Organizational Identity

The participants in the first workshop agreed on the following identity statement:

“Max Clean is a company that provides services in the facility services’ industry using cutting edge technology that aims to reach the industrial and residential sector hoping to optimize the client’s operation, working under international quality standards, offering excellence in its work, respecting the environment and creating value for the community”.

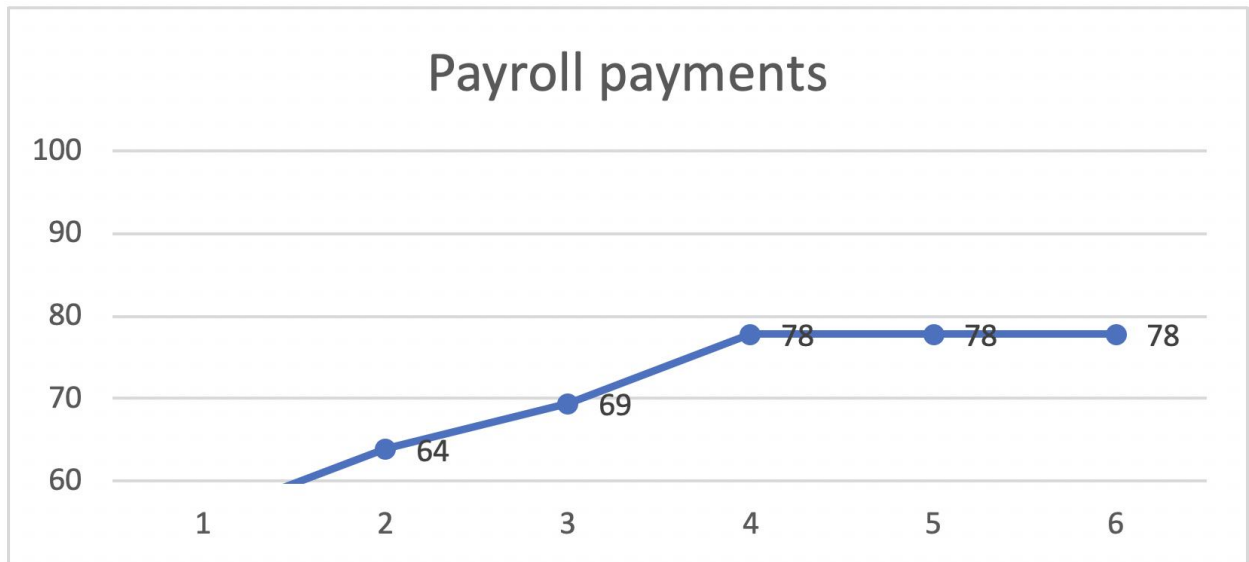
Determine the levels or recursive organization

In the second workshop participants agreed that Max Clean operated at three levels to manage its complexity: national, regional, and service as shown in Figure 6. The company organizes itself in six regional offices that each offer different types of services for industrial clients. Each of these services had special requirements that could not be undertaken by the same employees, so the company had to train its employees according to the type of service. In total there were 36 different industrial clients, with each serviced by a specific operational unit, as shown in Figure 6.



VSM Analysis

Preliminary data identified key diagnostic issues at the three levels of recursive organization, which were explored further during the VSM workshops. In general, it was clear that the same managers were operating at all 3 levels of recursion creating bottlenecks due to lack of time to effectively attend to them all (micro-managing), and there was a need to create a more focussed, flexible and decentralized structure, with clear leaders at each level with legitimacy and power (autonomy) to manage the branches.



During the VSM workshops, these issues were discussed, and decisions were made to address them in the short term. Participants also discussed the firm's adaptation and learning capabilities. Martinez-Lozada presented the full VSM diagnosis including two new stages for the STM: for design and implementation of sustainability strategies; and for delimiting the required capabilities for supporting sustainability strategy implementation. Six iterations of the methodology followed until the end of the project. At each iteration, a set of monitoring variables were examined to monitor implementation of the agreed changes.

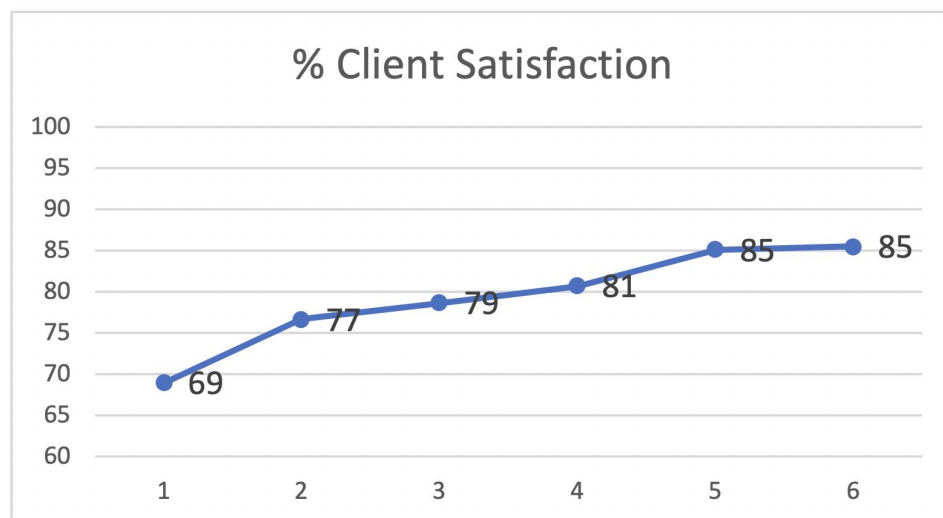
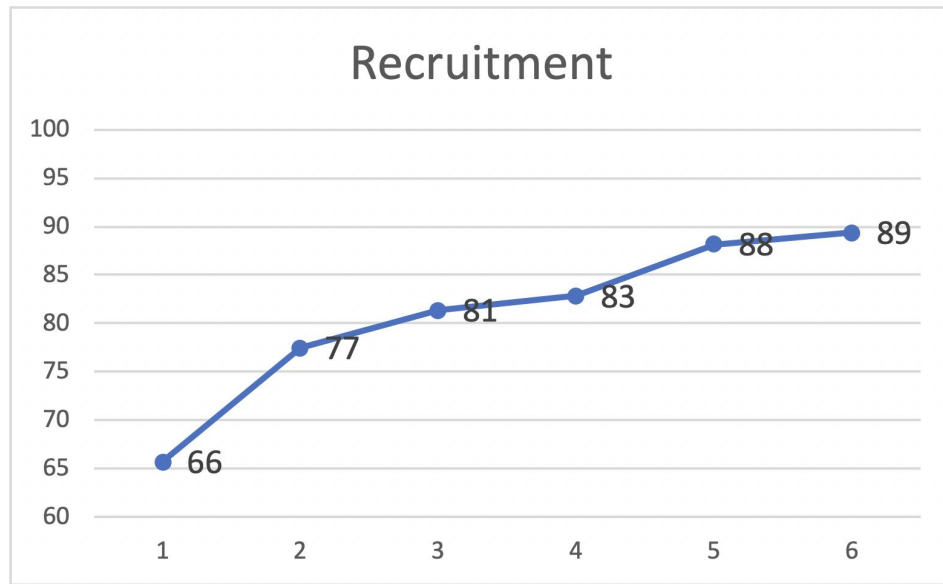
Agreement of changes (improvements) and evidenced effects on firm's performance

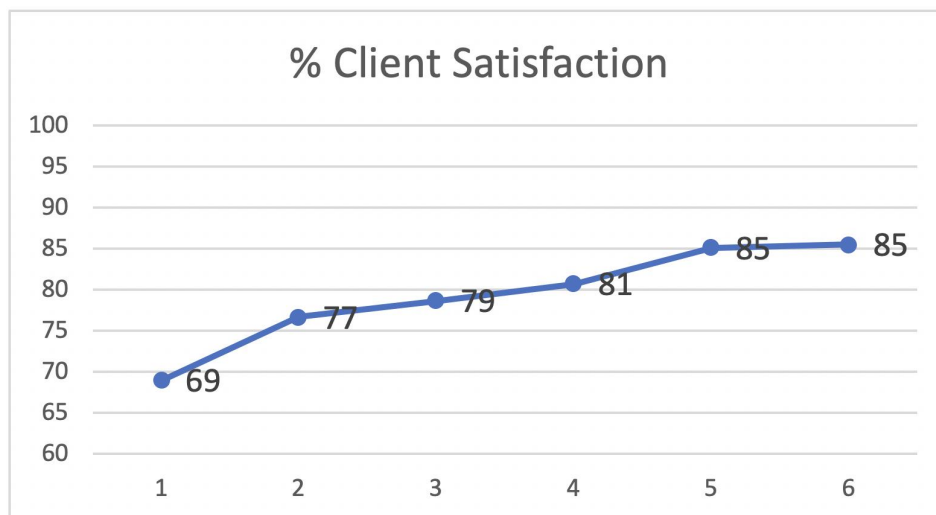
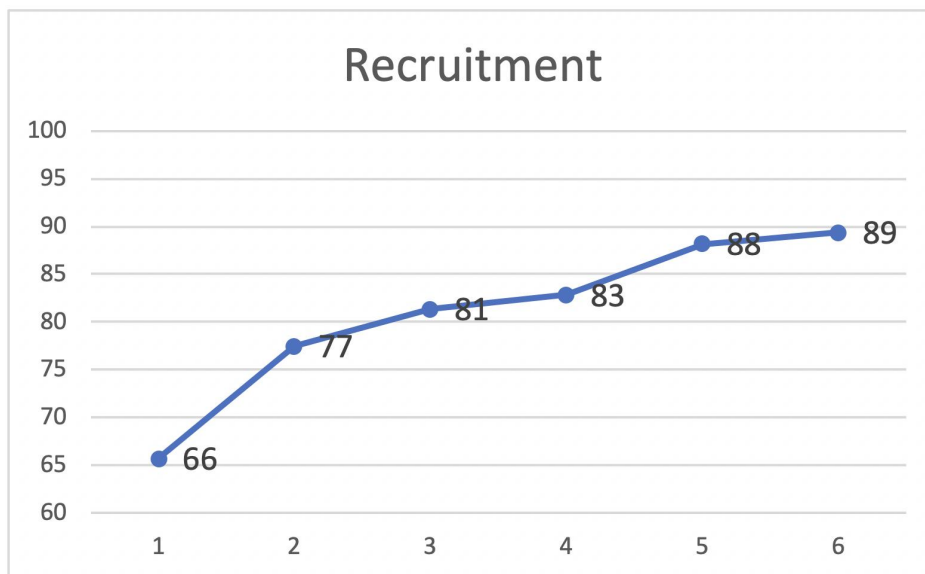
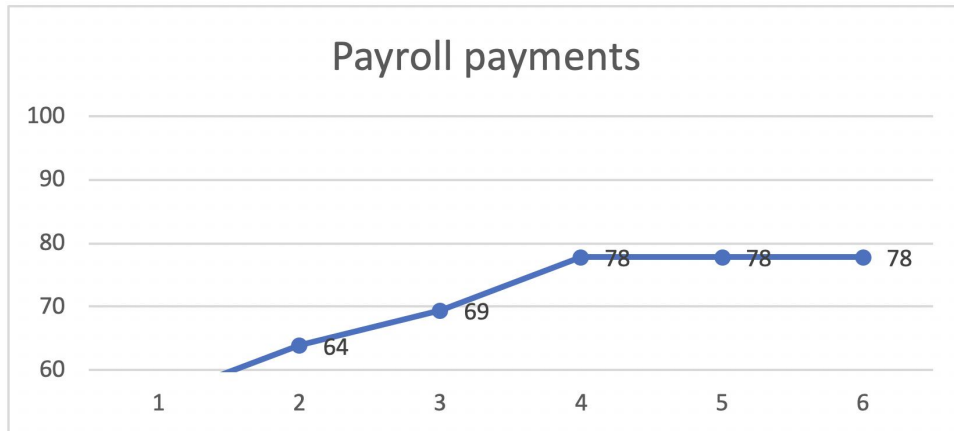
During the workshops, employees were able to propose and agree on action plans that were then implemented. At the national level, there was agreement to create an executive committee for strategy implementation, monitoring, defining organizational policies and developing a balanced scorecard. The regional level focused on improving system 3's managerial capacity by developing spaces for exchanging information and knowledge among Systems 1, and to develop management indicators of performance, and create more autonomy for the regions. At the service level, service leaders were appointed to assume system 5, 4 and 3 functions; develop performance indicators to track the performance of Systems 1; empower employees (especially in small services); and to assume more autonomy.

The implementation of these action plans was closely monitored during the intervention based on seven Key Performance Indicators (KPI). The company expected to see improvements in performance, complying with the firm's quality system, and it did: there were significant improvements in every index (see Figures 8-12). Five of the indices were related to service quality, including client satisfaction. In the social dimension, managers wanted to improve the quality of life of their employees (most of whom had low educational levels and lived in violent and insecure environments). They also wanted to improve

their environmental footprint, specifically the level of water contamination by improving procedures involving the use of chemicals.

Monitoring Performance

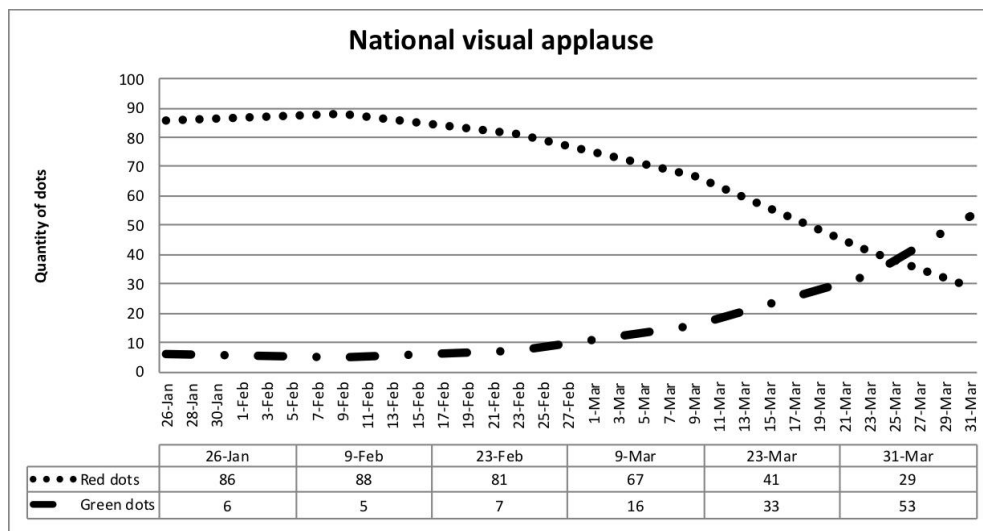




The company implemented an education project which offered training to 220 employees and their families (located in two cities) to increase their well-being. The firm also improved its environmental footprint: before the intervention it was using 1850 liters of chemicals per month, six months later it had dropped significantly to 1300 liters. It also began to consider using eco-friendly cleaning products.

Participants were also asked to re-evaluate their perceptions using the ‘visual applause’ technique. During each workshop, participants assigned diagnostic points green dots to show improvements) or red dots to indicate no change. Analysis showed that perceptions changed positively on almost all diagnostic points at every recursion level (see Figure 13 for national visual applause). The number of green dots indicate the problems that were solved during the time of the intervention and areas that exhibited exemplary performance .

The most significant improvement was at the national level, as the top management responded positively to the suggestions made and were able to immediately implement changes. This was despite the fact that it was hard for them to give up power and control over operations in order to give people at lower recursions the autonomy required.



This example shows how a VSM intervention can successfully work in a large company, by jointly agreeing on specific action plans to deal with both operational and socio environmental problems in a more sustainable way.

Conclusion: Learning from this application of the VSM

This case study confirms that the VSM is a useful tool for improving viability and sustainability in a company. It also shows the potential that VSM methodology offers for progressing into a more flexible structure to develop adaptive capabilities when trying to implement sustainability strategies. Importantly, it illustrates the powerful combination of a focused VSM diagnosis, KPI use, and the very short lag in the control loops (15 days), leading to an improved sustainability performance.

In summary, the implemented changes were as follows:

- All operational units at all levels of recursion were given more autonomy;
- Micro management was minimized;
- New KPIs were designed and measured regularly to create an infrastructure which focussed attention on important issues, provided accountability (so that senior management could monitor operational performance) and created learning loops so operational staff could assess and improve their performance;
- The 15 days period for meeting to review and make decisions resulted in very rapid and positive outcomes; and
- The need for a new strategic committee implemented System 4 at the national level.

The case study opens new avenues for researching sustainability measurement systems; and for studying the relationship between the dynamic capabilities framework and VSM theory. A next step would be to include these, and other KPIs, in the day to day management of the company and thus continue building social and environmental issues into the DNA of the company. Several avenues for further research are identified.

Frontier and future possible VSM developments

This chapter covers some of the more interesting aspects of the original VSM theory that have been developed recently: first, as a comprehensive underlying theory to clarify and put into practice the following: a) innovative self-organizing network structures, b) a clearer understanding of how to manage complexity in organizations c) new approaches to strategy, leadership and innovation, and d) practical ways for implementing sustainability strategies.

The pressing need for our economy to be less energy and resource intensive and to function within the constraints of a finite planet, opens new possibilities for further developing the VSM, in the directions suggested by the Viability & Sustainability approach [16,21–23; 26,4–61]. If System 5 places sustainability at the heart of its vision and values, then everything else falls into place with ease. The links between the Systems 1 and the outside world will require new metrics to embrace issues of resource use and pollution, and the flow of these metrics throughout the system would focus everyone's attention, in all subsystems, on the critical environmental issues rather than on a restricted financial set of objectives [24].

Understanding businesses as ecosystems has become popular in academia and in practice, and the VSM offers clear tools to map relationships with stakeholders at different levels of organized complexity. Areas such as Industrial Ecology are based on the concept of businesses striving to become more in balance with nature. The VSM can be used to design a cluster of businesses: each independent enterprise can be modeled as a Systems 1, and the Meta-system has the job of ensuring they work together as a coherent whole, working for their mutual benefit and doing less damage to the eco-systems with which they interact ([16], pp. 280–295).

Recent VSM research has confirmed the VSM can only be fully understood from a second order cybernetics perspective [14,16,17,21]. This, together with a continuously increasing number of VSM publications, demonstrates that practitioners have overcome old criticisms of the VSM (e.g. *Jackson's* [25] interpretation of the VSM as a functionalist approach, showing a difficult language, which creates substantial cognitive barriers). Espinosa [26] offers a comprehensive literature review on the VSM, and examples of innovative developments.

After 50 years, in all its many and varied applications, the VSM has provided insightful diagnosis and has directed organizational restructuring to create more effective organizations. The urgent need for highly effective businesses which are collaborative, less damaging to their environment, and based on individual flair and creativity has never been greater. After decades at the periphery, it is time for the VSM to take its position at centre stage.

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